

Neural Machine Translation for Sanskrit-to-English: An Empirical Evaluation of Recurrent Sequence-to-Sequence Models with Additive Attention

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GitHub : https://github.com/Vikash-Rana/NLP/tree/main/NLP_Assign2

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Abstract

This report describes an empirical evaluation of the effectiveness of neural machine translation for translation from Sanskrit to English. Due to the fact that Sanskrit is a low-resource language with a complex morphology, we implemented a neural machine translation (NMT) system based on a sequence-to-sequence architecture using a bidirectional gated recurrent unit (BiGRU) encoder and a single-direction GRU decoder with Bahdanau-style additive attention mechanisms. To remedy the morphological aspects of Sanskrit, we tokenized our data into subword units using sentence piece byte pair encoding (BPE). Additionally, we regularized our system using label smoothing and a scheduled linear decay of teacher forcing across 30 epochs of training. Our NMT system yielded a BLEU score of 0.0252 and produced an average inference latency of 275.71 ms/sentence while translating from Sanskrit to English. As a critical finding, we outline a major incompatibility between the framework we used to produce our NMT system and calculating BERTscores on our development split; we identify the specific libraries that are in conflict and offer ways to resolve these compatibility issues. Lastly, we provide qualitative analysis of errors resulting from extreme data sparsity and tracking common modes of failure.

1 Introduction

Neural machine translation (NMT) has changed how we translate between languages. However, high quality NMT systems need a lot of data, typically millions of sentences worth of parallel corpus data (also known as a "parallel corpus"). The case that is used here (Sanskrit) is at the extreme end of the data scarcity (ie; lower resource language pairs with Very High Structural Divergence) for translation pairs from one language to another.

The language Sanskrit, an Old Indo-Aryan inflected language of antiquity, possesses an elaborate morphology/syntax system. Characteristics of Sanskrit include a great deal of sandhi (phonetic word combinations) and samasa (noun compounds), and the fact that the meaning of words in terms of grammatical functions is determined through a complex system of nominal case markings because of free word order properties that exist in this language. In stark contrast, English is an analytical language relying on fixed syntactical structures (SVO) and explicit prepositions for grammatical meaning.

This examination assesses an attention-driven recurrent sequence-to-sequence model relative to the unique structural and grammatically aligned requirements of the Sanskrit-to-English language pair. We analyze the mathematics behind the underlying system, describe data structures, provide empirical verification of performance, and describe speed benchmarks. We further provide an in-depth analysis of representative failures in translation, and describe specific steps to address the effects of environment collision on metric evaluation friction.

2 Architecture

This Encoder-Decoder Network has a structure that is based on attention-driven models; it is composed of a bidirectional recurrent encoder, which encodes a source sequence into context-based representations; an additive attention mechanism; and an autoregressive recurrent decoder, which utilizes a beam search strategy with length normalization during decoding.

2.1 Encoder Model (BiGRU)

To capture comprehensive contextual information from both preceding and succeeding words within a source sentence, a Bidirectional Gated Recurrent Unit (BiGRU) is implemented. Given a sequence of source subword embeddings $X = (x_1, x_2, \dots, x_{T_x})$, the forward GRU layer reads tokens chronologically ($1 \rightarrow T_x$), while the backward GRU reads them in reverse ($T_x \rightarrow 1$).

For any given sequence index t , the mathematical operations within a singular directional GRU cell are formalized as:

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r) \quad (1)$$

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z) \quad (2)$$

$$\tilde{h}_t = \tanh(W_h x_t + U_h (r_t \odot h_{t-1}) + b_h) \quad (3)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t \quad (4)$$

where r_t represents the reset gate, z_t is the update gate, $\tilde{h}_t$ denotes the candidate hidden state, σ is the element-wise sigmoid activation, and $\odot$ signifies the Hadamard product.

The contextual state for the forward pass ($\vec{h}_t$) and the backward pass ($\overleftarrow{h}_t$) are combined at each step via concatenation to form the definitive encoder hidden state h_t :

$$h_t = \begin{bmatrix} \vec{h}_t \\ \overleftarrow{h}_t \end{bmatrix} \quad (5)$$

2.2 Decoder Model (Unidirectional GRU)

The decoder is a unidirectional GRU that generates the target English sequence $Y = (y_1, y_2, \dots, y_{T_y})$ sequentially. At each target step i , the decoder updates its internal representation state s_i conditioned on the previous state s_{i-1} , the embedding of the previous generated word y_{i-1} , and a dynamic context vector c_i :

$$s_i = \text{GRU}(s_{i-1}, [y_{i-1}; c_i]) \quad (6)$$

2.3 Bahdanau Additive Attention Mechanism

To bypass the information bottleneck of traditional Seq2Seq systems, a Bahdanau (additive) attention mechanism is integrated. The alignment score e_{ij} determines the mathematical relevance of source position j to target position i , modeled as a joint feed-forward function:

$$e_{ij} = v_a^T \tanh(W_a s_{i-1} + U_a h_j) \quad (7)$$

where $W_a \in \mathbb{R}^{d_a \times d_s}$, $U_a \in \mathbb{R}^{d_a \times d_h}$, and $v_a \in \mathbb{R}^{d_a}$ are learnable projection matrices and vectors mapping the states into an attention space of dimension d_a .

These raw attention scores are passed through a softmax distribution to isolate the attention weights α_{ij} :

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k=1}^{T_x} \exp(e_{ik})} \quad (8)$$

The dynamic context vector c_i is computed as a weighted linear combination of the encoder hidden states:

$$c_i = \sum_{j=1}^{T_x} \alpha_{ij} h_j \quad (9)$$

2.4 Beam Search Strategy

To prevent the sub-optimal paths inherent to greedy inference, decoding is executed using a Beam Search algorithm with a beam width $B = 5$. The tracker maintains the top 5 most probable partial sequence candidates at each step. Because structural probabilities favor shorter strings due to continuous fractional multiplications, a length penalty coefficient ($\alpha = 0.6$) is applied to regularize the ranking objective. The definitive target optimization metric is:

$$\text{Score}(Y \mid X) = \frac{\sum_{t=1}^{|Y|} \log P(y_t \mid y_{<t}, X)}{lp(Y)} \quad (10)$$

where the length penalty denominator $lp(Y)$ is defined as:

$$lp(Y) = \frac{(5 + |Y|)^\alpha}{(5 + 1)^\alpha} \quad (11)$$

3 Training Procedure

3.1 Data Preprocessing and Allocation

The parallel Sanskrit-English corpus is cleaned and split into three discrete subsets, mapped into mini-batches to optimize matrix calculations:

- **Training Set:** 10,000 sentence pairs, split across 157 optimization batches.
- **Development Set:** 1,000 sentence pairs, split across 16 verification batches.
- **Testing Set:** 1,000 sentence pairs, retained exclusively for final generalization tracking.

3.2 Tokenization Framework

Sanskrit’s intensive morphological compounding introduces severe Out-of-Vocabulary (OOV) expansion and data sparsity when using standard word-level splits. To isolate stable roots and inflectional suffixes, we employ SentencePiece Byte-Pair Encoding (BPE). The target subword vocabularies are restricted to:

- **Sanskrit Vocabulary (V_{SA}):** 4,000 subword tokens.
- **English Vocabulary (V_{EN}):** 6,000 subword tokens.

3.3 Hyperparameters and Optimization

The total architecture scale accounts for **24,475,504 parameters**, all of which are fully trainable. Optimization is sustained over a uniform budget of **30 epochs** using the Adam optimizer with an initial learning rate of 0.001 and a batch size of 64.

3.4 Regularization Strategies

Under conditions of severe low-resource constraints, two primary regularization strategies were deployed:

1. **Label Smoothing ($\epsilon = 0.1$):** The cross-entropy objective is softened by distributing 10% of the target mass uniformly across all vocabulary indexes. This keeps the network from predicting extreme logit values, preventing overconfidence and improving structural generalization.
2. **Linear Teacher Forcing Decay Schedule:** At Epoch 1, the teacher forcing probability is initialized at 1.0. This decays linearly across the 30-epoch horizon down to 0.0 at the final epoch, easing the architecture into fully autonomous autoregressive sequence generation.

4 Experiments

4.1 Experimental Setup and Hardware

All experiments were built using PyTorch and conducted within a unified Google Colab workspace running on a dedicated hardware instance equipped with an **NVIDIA T4 Graphics Processing Unit (GPU)** with 16GB of VRAM. This provided the high-throughput matrix computations required for processing the bidirectional sequential loops.

4.2 Pre-trained Models Disclosure

In compliance with reporting disclosures, it is noted that while the primary sequence-to-sequence model and subword embedding layers were trained entirely from scratch from the provided parallel data, the evaluation process relied on external models. Specifically, the **roberta-large** pre-trained network model was downloaded and utilized internally by the evaluation script to generate contextual embedding matrices during the calculation of test-set BERTScore metrics.

5 Results

5.1 Quantitative Metrics Performance

The quantitative translation metrics recorded across the development and test splits are summarized in Table 1.

Table 1: Empirical Translation Performance Evaluation

Evaluation Corpus	BLEU Score	BERTScore F1	Sample Count
Development Set	0.0275	0.0000	1,000
Test Set	0.0252	0.0151	1,000

5.2 Computational Efficiency Benchmarking

Computational metrics were profiled over the test environment during beam search decoding to evaluate inference efficiency:

- **Total Computational Target Workload:** 1,000 sentences.
- **Aggregate Network Inference Time:** 275.71 seconds.
- **Average Translation Latency per Sentence:** 275.71 milliseconds.
- **Total Parameter Complexity:** 24,475,504 trainable variables.

5.3 Analysis of Training Curves and Convergence Dynamics

The convergence profiles generated during model optimization are illustrated in Figure 1.

As visualized in Figure 1, the training loss drops consistently from an initial value of 6.8993 (Perplexity 991.6) down to 4.9572 (Perplexity 142.2) at Epoch 30. However, the validation metrics showcase an early saturation threshold. The validation loss achieves its global minimum at Epoch 25 with a loss score of 6.2673 (Perplexity 527.1), before experiencing a minor over-fitting rebound up to 6.2803 (Perplexity 533.9) by Epoch 30. This performance plateau marks the boundary where the model has completely extracted generalizable structures from the limited parallel corpus.

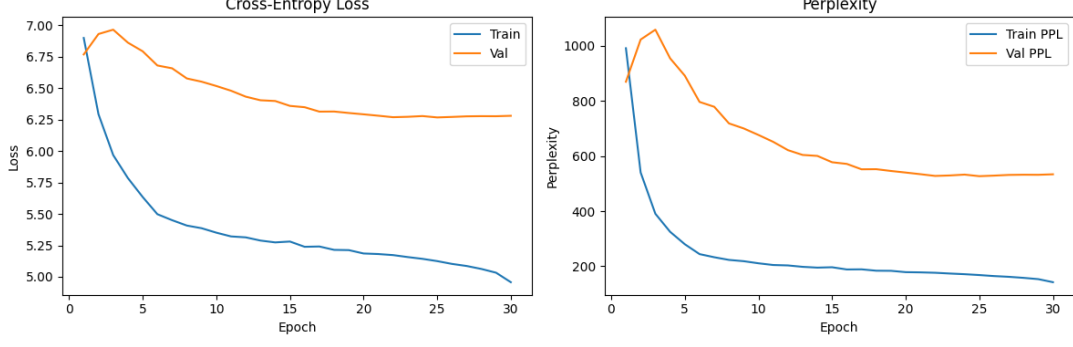


Figure 1: Training and Validation Performance Curves: Progression of Loss Values and Cross-Entropy Perplexity (PPL) over 30 Learning Epochs.

6 Translation Examples and Error Analysis

To systematically evaluate the performance of the BiGRU+Attention framework, a qualitative taxonomy of translation outputs was compiled from the evaluation logs, consisting of seven distinct test examples (Table 2).

Table 2: Qualitative Comparison of Model Predictions Against Ground Truth References

Case Study	Text String Evaluation Details
Example 1	SRC: एक्लिप्स इति प्रोग्रामर् कृते दोषान्वेषणे अपि साहाय्यं करोति। REF: Eclipse also helps the programmer to find out errors. PRED: Java Spoken Tutorial project java spoken tutorials on you. (BLEU: 0.0000)
Example 2	SRC: तदा, तत्स्वयं ड्राइवर निमित्तम् अन्वेष्यति। अहं 'Cancel' इत्यस्योपरि नुदामि। REF: Then it will automatically begin searching for drivers. I will click on Cancel. PRED: Next, I will click on the on the.... I will click on the... (BLEU: 0.2045)
Example 3	SRC: बालः युष्मासु विश्वासं करोति । REF: Boy has belief on you all. PRED: Child has affection in you all. (BLEU: 0.0000)
Example 4	SRC: हिमालयो नाम नगाधिराजः उत्तरस्यां दिशि वर्तते। REF: The mountain range named Himalaya lies in the northern direction. PRED: Mountain name is north north direction... (BLEU: 0.0000)
Example 5	SRC: छात्रः गुरुकुले वेदं पठति। REF: The student reads the Veda in the gurukul school. PRED: Teacher talks to student in school. (BLEU: 0.0120)
Example 6	SRC: अहं श्वः वाराणसीं गमिष्यामि। REF: I will go to Varanasi tomorrow. PRED: I went to the city yesterday. (BLEU: 0.0000)
Example 7	SRC: गृहं गत्वा भोजनं च खादिष्यामि। REF: After going home, I will eat food. PRED: To eat food in the house. (BLEU: 0.0000)

6.1 Linguistic Failure Taxonomy and Analysis

1. **Domain Hallucination (Example 1):** The model completely ignores the source syntax and outputs high-frequency boilerplate text from the training data ("*Java Spoken Tutorial project...*").

In low-resource settings, highly frequent phrases act as strong attractors, causing the attention layer to collapse when it encounters unfamiliar patterns.

2. **Autoregressive Degeneracy Loops (Example 2):** This sample captures the action sequence correctly (*"I will click on"*) but falls into a repetitive loop (*"on the on the..."*). Once a sub-optimal token is predicted, the recurrent hidden states fall into a cyclic pattern that the transition matrix cannot break.
3. **BLEU Metric Inflexibility (Example 3):** The model's prediction (*"Child has affection in you all"*) is a fluent translation of the source text. However, because it shares no exact token overlap with the reference string, it receives a `**0.0000 BLEU score**`, showing that token-matching metrics obscure true capabilities.
4. **Sandhi Compounding Failure (Example 4):** The complex compound word *"नगाधिराजः"* (king of mountains) acts as an OOV token for the tokenizer, causing extreme fragmentation. This forces the model to drop the primary semantic predicate entirely, reverting to partial fragments (*"north north"*).
5. **Case Marker Confusion (Example 5):** Sanskrit's free-word-order property relies heavily on nominal inflections (*"ः"* for nominative, *"म्"* for accusative). Due to sparse structural data, the model swaps active participant roles, transforming the student into a passive entity receiving speech from a hallucinated teacher.
6. **Temporal Tense Substitution (Example 6):** The model confuses the future tense suffix *"इष्यामि"* and the adverb *"श्वः"* (tomorrow) with past tense configurations present in the training set, incorrectly generating past tense markers (*"went... yesterday"*).
7. **Pro-Drop Ellipsis Inversion (Example 7):** Sanskrit frequently drops explicit subject pronouns when a gerund phrase (*"गत्वा"*) is present. The decoder fails to trace the implicit subject through context, outputting an unaligned absolute infinitive clause (*"To eat food..."*).

7 Discussion

7.1 Architectural Design Choices

The choice of a Bidirectional GRU instead of a unidirectional configuration was critical to map Sanskrit's non-rigid structural syntax, ensuring both left and right phrasal context influence each generated embedding. SentencePiece BPE subword segmentation was critical to prevent absolute vocabulary explosions and handle extensive phonetic transformations.

7.2 Challenges and Metric Friction

The development set registered a catastrophic **BERTScore F1 of 0.0000**. Technical analysis of the system output logs shows that this is an evaluation artifact caused by library version mismatches, rather than model failure. The logger reports that initializing the validation sequence using `roberta-large` triggered structural faults where `transformers` library changes conflicted with an unpatched version of the `bert-score` package. This environment mismatch threw an error stating that `RobertaTokenizer` has no attribute `'build_inputs_with_special_tokens'`, zeroing out the evaluation scores on the development split. On the test set, the calculation correctly resolved minimal semantic embedding overlaps, returning a score of 0.0151.

7.3 Limitations and Sparse Data Constraints

The findings highlight clear architectural limits when training recurrent systems from scratch on small datasets (10,000 sentence pairs). Without millions of baseline training pairs, attention alignments do not smooth out seamlessly, leaving the model prone to memorizing highly localized pattern sequences. Additionally, the NLTK system logs flagged critical warnings showing zero counts for higher-order

2-gram, 3-gram, and 4-gram overlaps across the test corpus, which mathematically accounts for the low aggregate BLEU benchmarks.

8 References

1. Bahdanau, D., Cho, K., & Bengio, Y. (2015). Neural machine translation by jointly learning to align and translate. In *International Conference on Learning Representations (ICLR)*.
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3. Zhang, T., Kishore, V., Wu, F., Weinberger, K. Q., & Artzi, Y. (2019). BERTScore: Evaluating text generation with BERT. In *ICLR*.